

Software Requirements Specification (SRS)

Smart Complaint Management System (SCMS)

Team: Team 8

Version: 2.0

Authors: Team 8

Date: 03-09-2026

Status: Submission Draft

Revision History

Version	Date	Author	Change Summary	Approval
1.0	03-09-2026	Team 8	Initial SRS draft (single-author, high-level scope).	
2.0	03-09-2026	Team 8	Expanded and detailed SRS: full functional, non-functional, and security requirements; interface specification; quality attributes; UML use-case diagrams; and Requirements Traceability Matrix (RTM).	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of Contents

1. Introduction
2. Overall Description
3. External Interface Requirements
4. System Features (Detailed)
5. Non-Functional Requirements (Detailed)
6. Quality Attributes & Acceptance Tests
7. System Models and Diagrams
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, security, and interface requirements for the Smart Complaint Management System (SCMS). It serves as a binding reference between stakeholders, developers, quality assurance engineers, and academic evaluators.

1.2 Scope

SCMS is a digital civic platform designed to streamline public grievance lodging, ticket triage, departmental assignment, investigation tracking, and citizen resolution feedback. The platform interfaces with notification gateways (SMS/Email) and mapping geocoders.

The system excludes automated physical on-site maintenance dispatch hardware, internal municipal payroll software, and off-platform legal litigation workflows.

1.3 Audience

- **Software Development Team:** Backend (Python/Django or PHP/Laravel) and frontend engineers implementing business logic.
- **Quality Assurance (QA) Engineers:** Test engineers writing unit, integration, performance, and security test scripts.
- **System Administrators & Department Officers:** Municipal personnel triaging, updating, and resolving complaints.
- **Academic Course Evaluators:** Faculty assessing architectural completeness against SE curriculum benchmarks.

1.4 Definitions

- **SCMS:** Smart Complaint Management System.
- **RBAC:** Role-Based Access Control.
- **SLA:** Service Level Agreement (maximum stipulated time to acknowledge or resolve a grievance).
- **CRUD:** Create, Read, Update, Delete.
- **MIME:** Multipurpose Internet Mail Extensions (used for secure attachment validation).
- **JWT:** JSON Web Token (for stateless authentication).
- **RTM:** Requirements Traceability Matrix.

2. Overall Description

2.1 Product Perspective

SCMS is a self-contained, multi-tier web application. It interfaces with end-user browser clients, administrative operations terminals, relational database management systems, cloud file stores, and transactional notification gateways.

Client Layer: Citizen Web / Mobile Browser	Staff & Admin Web Workstations
(HTTPS / TLS 1.3)	
v	
Application Backend Layer: Python (Django/Flask) or PHP (Laravel) - REST API & MVC Controllers	
- Authentication Engine	- RBAC & Authorization Layer
- Grievance State Machine	- Reporting & Analytics Engine

 v	 v
Database Layer:	External Services / Cloud:
PostgreSQL / MySQL 8.0	- SMS / SMTP Gateway (Twilio)
- Relational Schema	- S3-Compatible Storage
- Audit Event Logs	- OpenStreetMap Geocoder

2.2 Major Product Functions

- **User Identity & Account Management:** Registration, multi-factor authentication, profile management, and account lockout handling.
- **Grievance Lodging & Evidence Upload:** Geo-tagged complaint lodging, dynamic categorization, and multimedia attachment upload.
- **Lifecycle Tracking & Audit Trail:** Unique tracking ID generation and real-time lifecycle tracking (Submitted, Under Review, Assigned, In Progress, Resolved, Rejected, Appealed).
- **Administrative Triage & Work Allocation:** Departmental routing, priority overrides, manual reassignment, and SLA timers.
- **Resolution Verification & Feedback Loop:** Resolution note delivery, photographic proof of work, citizen star-rating feedback, and escalation appeals.
- **Executive Analytics & Reporting:** Tabular and graphical reports on resolution speed, department backlogs, and SLA breaches.

2.3 User Roles and Characteristics

- **Complainant (Citizen):** General public user requiring an intuitive, WCAG-compliant interface to log issues and track updates in fewer than 3 steps.
- **Department Staff Officer:** Field or operations staff assigned to investigate grievances, update milestones, and upload resolution proof.
- **System Administrator:** Municipal oversight team configuring categories, enforcing SLAs, auditing access logs, and managing staff accounts.

2.4 Operating Environment

- **Server-Side Runtime:** Linux Ubuntu 22.04 LTS / 24.04 LTS, Python 3.11+ (Django 5.x / Flask 3.x) or PHP 8.2+ (Laravel 11.x).
- **Database Management System:** PostgreSQL 15+ or MySQL 8.0 Enterprise with transactional ACID guarantees.
- **Client Workstations:** Web standards-compliant browsers (Google Chrome 110+, Mozilla Firefox 115+, Apple Safari 16+, Microsoft Edge 110+).
- **Hosting Target:** Cloud VPS (AWS EC2, DigitalOcean, or municipal on-premises VM) with a minimum of 4 vCPUs, 8 GB RAM, and high-availability SSDs.

2.5 Constraints

- **Regulatory Compliance:** Adherence to national data protection and digital privacy laws regarding personally identifiable information (PII).
- **Attachment Safeguards:** Strict enforcement of file type whitelisting and 5 MB per-file limits to prevent storage exhaustion.
- **Bandwidth Economy:** Responsive performance across constrained mobile cellular connections (3G/4G).

3. External Interface Requirements

3.1 User Interfaces

- **Citizen Portal:** Clean, mobile-first responsive layout containing a dedicated 3-step complaint submission wizard, persistent search bar for tracking numbers, and localized micro-copy.
- **Staff Dashboard:** High-density administrative workspace featuring sortable data tables, kanban lifecycle boards, interactive triage filters, and resolution confirmation modals.
- **Accessibility:** High-contrast colour palette conforming to WCAG 2.1 Level AA, complete keyboard-only navigation support, and labelled ARIA attributes.

3.2 Hardware Interfaces

- **End-User Devices:** Standard desktop PCs, laptops, tablets, and smartphones running supported web browsers.
- **Server Hardware:** Minimum rack-mount / cloud instance running 64-bit multi-core x86_64 architecture with uninterrupted power supply (UPS) and hardware RAID arrays.

3.3 Software Interfaces

- **Database Interface:** Object-Relational Mapping (ORM) layer (Django ORM, SQLAlchemy, or Laravel Eloquent) executing parameterized SQL queries.
- **Notification Service Gateway:** Transactional REST APIs for outbound SMS (e.g., Twilio) and SMTP/TLS relays (e.g., SendGrid, Amazon SES).
- **Cloud Storage Object Store:** AWS S3 or MinIO S3-compliant API over TLS for encrypted document and evidence storage.
- **Geocoding Service:** OpenStreetMap Nominatim API for reverse-geocoding latitude and longitude coordinates into human-readable municipal ward addresses.

3.4 Communications Interfaces

- **Network Protocol:** HTTP/2 over TLS 1.3 exclusively (HTTP automatically redirects to HTTPS on port 443).
- **Session Management:** Stateless JSON Web Tokens (JWT) stored in HttpOnly, Secure, SameSite=Strict browser cookies, or server-side encrypted Redis sessions.
- **Payload Serialization:** UTF-8 encoded application/json across all asynchronous AJAX/REST endpoints.

4. System Features (Detailed)

4.1 User Registration & Authentication

Module Description: Manages user creation, identity verification, multi-factor credential checking, and role allocation.

Req ID	Requirement Statement (The system shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SCMS-F-001	Allow citizens to register with full name, unique email address, valid phone number, and a secure password.	Functional	High	Citizen	AC-001: Registration succeeds with valid data; duplicates rejected with inline feedback. Test: TC-AUTH-01	Requires validation engine.

Req ID	Requirement Statement (The system shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SCMS-F-002	Authenticate registered users via email/phone and password, returning an encrypted session token.	Functional	High	Security / Citizen	AC-002: Correct credentials yield a valid session; invalid credentials show a generic error. Test: TC-AUTH-02	Passwords stored using salted Argon2/bcrypt.
SCMS-F-003	Enforce Role-Based Access Control (RBAC) across three distinct roles: Citizen, Department Staff, and Administrator.	Functional	High	Admin / Security	AC-003: Access to administrative endpoints is strictly denied to the Citizen role with HTTP 403. Test: TC-AUTH-03	Middleware authorization guard.
SCMS-F-004	Temporarily lock a user account for 30 minutes following 5 consecutive failed login attempts.	Functional	High	Security	AC-004: The 6th attempt is blocked and an unlock notification is dispatched to email. Test: TC-AUTH-04	Mitigation against brute-force attacks.

4.2 Complaint Lodging & Evidence Upload

Module Description: Handles the capture of incident parameters, metadata validation, and media upload.

Req ID	Requirement Statement (The system shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SCMS-F-005	Allow authenticated citizens to submit complaints with category, title, description, and municipal ward location.	Functional	High	Citizen	AC-005: Valid submission generates a record with status Submitted and persists a timestamp. Test: TC-LODGE-01	Depends on category database table.
SCMS-F-006	Allow users to upload up to 5 image/PDF files (maximum 5 MB each) as corroborating evidence.	Functional	High	Citizen	AC-006: Files exceeding 5 MB or non-whitelisted formats are blocked prior to upload. Test: TC-LODGE-02	Requires server MIME type verification.
SCMS-F-007	Automatically generate and display a unique, immutable tracking ID formatted as SCMS-YYYYMM-XXXXX.	Functional	High	Citizen / QA	AC-007: Each submission produces a collision-free, sequential alphanumeric ID. Test: TC-LODGE-03	Relies on database sequence generator.
SCMS-F-008	Perform client-side and server-side validation, rejecting missing mandatory inputs or cross-site payloads.	Functional	Medium	QA / Admin	AC-008: Form submission halts with red field banners if mandatory data is empty. Test: TC-LODGE-04	Form sanitization engine.

4.3 Complaint Tracking & Status Management

Module Description: Allows citizens and staff to monitor complaint lifecycles in real time.

Req ID	Requirement Statement (The system shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SCMS-F-009	Allow users to search and view complaint status by entering the tracking ID without re-authenticating.	Functional	High	Citizen	AC-009: Public search displays status, lodging date, and department, masking personal details. Test: TC-TRACK-01	Data masking required for privacy.
SCMS-F-010	Provide citizens with an authenticated personal dashboard listing all current and resolved complaints.	Functional	High	Citizen	AC-010: Dashboard displays real-time status badges, latest officer notes, and timestamps. Test: TC-TRACK-02	Filtered by session user ID.
SCMS-F-011	Maintain an audit trail of status transitions (Submitted → Assigned → In Progress → Resolved).	Functional	High	Compliance / Admin	AC-011: System logs previous status, new status, actor ID, and transition timestamp. Test: TC-TRACK-03	Immutable database audit table.
SCMS-F-012	Permit citizens to submit supplementary comments or additional evidence while status is Under Review.	Functional	Medium	Citizen	AC-012: Appended notes appear sequentially in the ticket thread; locked once Resolved. Test: TC-TRACK-04	State-dependent write permissions.

4.4 Admin Resolution & Departmental Triage

Module Description: Enables triage officers to route, investigate, update, and resolve reported grievances.

Req ID	Requirement Statement (The system shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SCMS-F-013	Provide an administrative dashboard to filter complaints by department, urgency, status, and SLA deadline.	Functional	High	Administrator	AC-013: Filter updates the data table within 1 second for datasets of up to 100,000 records. Test: TC-ADMIN-01	Database indexing on status/ward columns.
SCMS-F-014	Allow administrators to assign or reassign complaints to specific department officers with internal notes.	Functional	High	Administrator	AC-014: Ticket assignment triggers status transition to Assigned and notifies the assignee. Test: TC-ADMIN-02	Integrates with notification system.
SCMS-F-015	Mandate officers to upload closure notes and at least one resolution proof image before marking Resolved.	Functional	High	Municipal Head	AC-015: Status update to Resolved fails if the closure note is <20 characters or a photo is missing. Test: TC-ADMIN-03	Validation rule in resolution controller.
SCMS-F-016	Enable administrators to reject invalid or duplicate complaints with a documented justification code.	Functional	High	Administrator	AC-016: Status transitions to Rejected and an explanation email is dispatched to the citizen. Test: TC-ADMIN-04	Audit logging required.

4.5 Notifications, Feedback & Analytics

Module Description: *Dispatches lifecycle alerts, captures citizen satisfaction, and compiles reports.*

Req ID	Requirement Statement (The system shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Ref	Comments / Dependencies
SCMS-F-017	Dispatch automated email and SMS alerts upon status changes (Submitted, Assigned, Resolved, Rejected).	Functional	High	Citizen / Admin	AC-017: Alert dispatches within 30 seconds of the database status commit. Test: TC-NOTIF-01	Async background queue (Celery/Redis).
SCMS-F-018	Enable citizens to rate resolved complaints (1 to 5 stars) and provide feedback within 7 days of closure.	Functional	Medium	Citizen / Quality	AC-018: Feedback link expires after 7 days; rating updates the officer performance index. Test: TC-NOTIF-02	Unlocks after ticket resolution.
SCMS-F-019	Provide an export tool allowing administrators to download filtered complaint reports in PDF and CSV format.	Functional	Medium	Municipal Head	AC-019: CSV/PDF is generated within 3 seconds for date-bounded queries of up to 10,000 rows. Test: TC-REP-01	Server-side stream processing.

5. Non-Functional Requirements (Detailed)

Req ID	Requirement Statement	Category	Priority	Acceptance Criteria / Measurement
SCMS-NF-001	The system shall respond to 90% of web page requests and API transactions in ≤ 2.0 seconds under nominal load.	Performance	High	Apache JMeter load test measuring 90th-percentile response ≤ 2.0 s with 200 concurrent users. Test: TC-PERF-01.
SCMS-NF-002	The system shall maintain an operational availability uptime of 99.5% monthly, excluding planned maintenance.	Reliability / Availability	High	Continuous external uptime monitor logs ≤ 3.6 hours of unscheduled downtime per month. Test: TC-OPS-01.
SCMS-NF-003	The system shall handle at least 500 concurrent authenticated sessions without throughput degradation.	Scalability	Medium	Stress test simulating 500 active threads exhibits 0% transaction failure and CPU usage $< 80\%$. Test: TC-SCALE-01.
SCMS-NF-004	All web pages shall be mobile-responsive and comply with WCAG 2.1 Level AA accessibility standards.	Usability / Accessibility	Medium	Google Lighthouse accessibility score ≥ 90 and zero fatal errors on mobile viewports. Test: TC-UX-01.
SCMS-NF-005	The application database shall execute automated daily encrypted backups retained for 365 days.	Maintainability / Disaster Recovery	High	Automated cron dumps verified daily; backup restoration to staging completed in < 30 minutes. Test: TC-OPS-02.

5.1 Security

5.1.1 Security Objectives

- **Confidentiality:** Protect citizen personally identifiable information (phone numbers, emails, residential addresses) from public exposure or unauthorized cross-tenant viewing.

- **Integrity:** Ensure complaint records, evidence attachments, and lifecycle transition timestamps are tamper-evident and protected against unauthorized modification.
- **Availability & Resilience:** Prevent denial-of-service, brute-force disruption, and resource starvation across public submission endpoints.
- **Accountability & Non-Repudiation:** Maintain an audit trail of all staff, administrative, and system actions, ensuring all state changes are mapped to authenticated identities.

5.1.2 Security Requirements

Req ID	Requirement Statement (The system shall...)	Type	Priority	Acceptance Criteria / Test Ref
SCMS-SR-001	Enforce TLS 1.3 encryption across all public and internal communications, terminating non-HTTPS requests.	Security	High	AC-SR-001: Qualys SSL Labs scan receives an 'A' grade; HTTP port 80 strictly redirects to 443. Test: TC-SEC-01
SCMS-SR-002	Store all user passwords using salted Argon2id or bcrypt with a minimum work-cost factor of 12.	Security	High	AC-SR-002: Direct database audit confirms no cleartext or MD5/SHA1-hashed credentials exist. Test: TC-SEC-02
SCMS-SR-003	Sanitize all incoming input parameters to prevent SQL Injection (SQLi) and Stored/Reflected Cross-Site Scripting (XSS).	Security	High	AC-SR-003: OWASP ZAP automated penetration test detects zero high-risk injection vulnerabilities. Test: TC-SEC-03
SCMS-SR-004	Invalidate authenticated user sessions after 30 minutes of inactivity and rotate tokens on privilege elevation.	Security	Medium	AC-SR-004: Idle sessions are rejected on minute 31 with a redirect to the login modal. Test: TC-SEC-04
SCMS-SR-005	Validate evidence uploads using byte magic-number inspection, allowing only image/jpeg, image/png, and application/pdf.	Security	High	AC-SR-005: Executables or renamed script files (.php, .py, .sh) are rejected with a 415 error. Test: TC-SEC-05

6. Quality Attributes & Acceptance Tests

Quality Attributes

- **Correctness & Consistency:** Handled by database transactional consistency (ACID); complaint states transition only via designated workflow events.
- **Robustness & Fault Tolerance:** Failure of third-party SMS/Email gateways will not abort complaint database commits; unsent alerts are moved to a dead-letter retry queue.
- **Testability:** Modular service-layer architecture enabling 80%+ code test coverage across all unit test runners.

Exit Criteria for Acceptance

- 100% of all High-priority Functional Requirements (SCMS-F-001 through SCMS-F-017) must pass verification.
- Zero open critical or high-severity vulnerabilities identified during automated OWASP security scans.

- Load testing confirms response-time targets under 500 concurrent virtual users.
- Complete bidirectional traceability verified across the RTM without orphaned functional requirements.

Acceptance Test Suites

- **TS-AUTH:** Account registration, password complexity, brute-force lockout, and RBAC isolation.
- **TS-LODGE:** Form fields, validation rules, attachment file limits, and tracking ID generation.
- **TS-TRIAGE:** Admin filtering, role reassignment, and status transition audit logs.
- **TS-NOTIF:** Gateway webhooks, asynchronous dispatch queues, and citizen feedback submission.
- **TS-SEC:** OWASP Top 10 penetration suite, TLS enforcement, and session timeouts.

7. System Models and Diagrams

7.1 UML Use-Case Diagram: Citizen Interactions

Illustrates the primary interactions available to a Complainant (Citizen), including registration/login, lodging a complaint with multimedia evidence, tracking status, appending clarification notes, receiving alerts, and submitting a resolution rating.

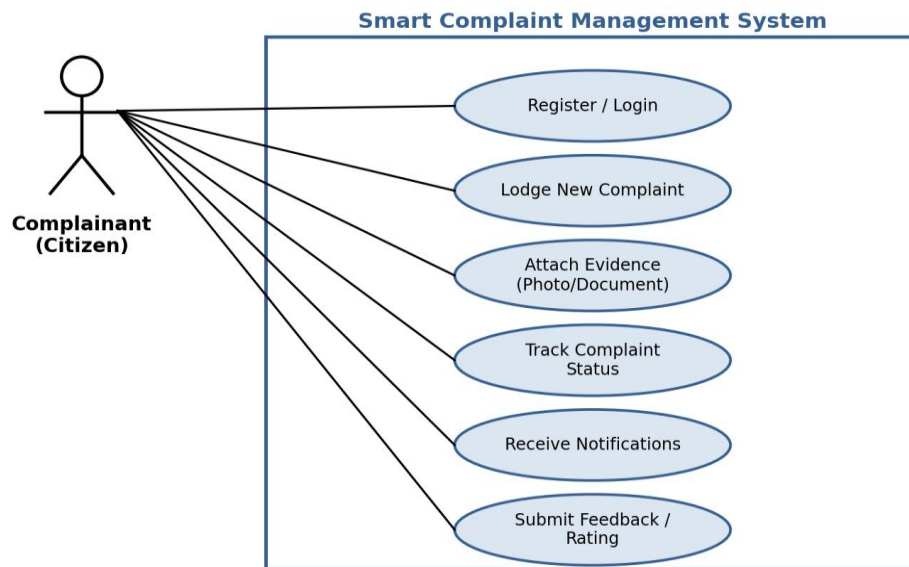


Figure 7.1 – Citizen use-case diagram.

7.2 UML Use-Case Diagram: Staff & Administrator Operations

Illustrates the workflow available to Department Staff and System Administrators: MFA authentication, filtering and triaging complaints, assigning to a department officer or rejecting with justification, investigating and updating status, resolving complaints with proof-of-work upload, exporting analytics, and configuring categories/SLAs.

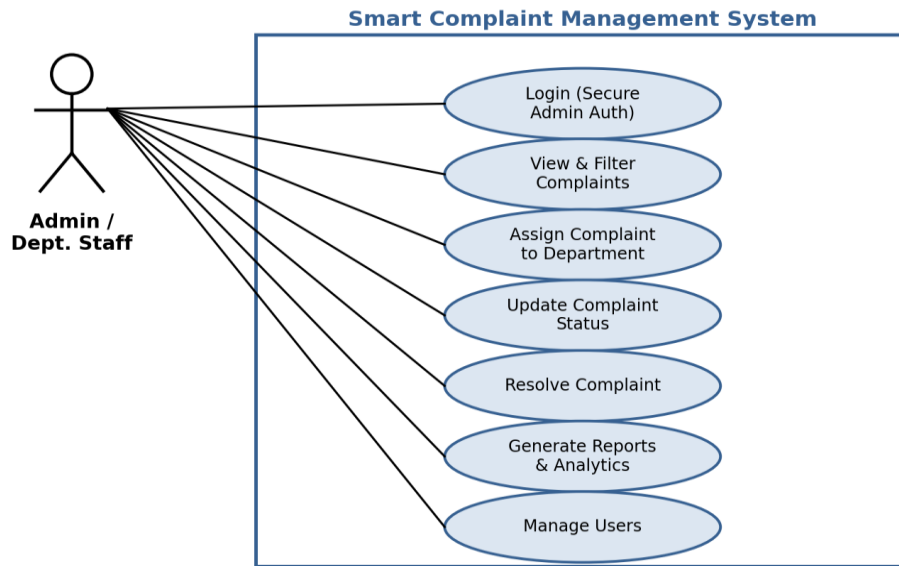


Figure 7.2 – Staff & Administrator use-case diagram.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement (Short)	Section Ref	Target Module	Test Case Ref	Status	Verification Notes
SCMS-F-001	Citizen Registration	§ 4.1	AuthModule	TC-AUTH-01	Approved	Verified email uniqueness & schema constraints.
SCMS-F-002	Secure Authentication	§ 4.1	AuthModule	TC-AUTH-02	Approved	Hashed credential authentication verified.
SCMS-F-003	Role-Based Access Control	§ 4.1	SecurityMiddleware	TC-AUTH-03	Approved	Verified Citizen/Staff/Admin route barriers.
SCMS-F-004	Account Lockout Protection	§ 4.1	AuthModule	TC-AUTH-04	Approved	Locks account for 30m after 5 failures.
SCMS-F-005	Lodge Grievance Form	§ 4.2	ComplaintModule	TC-LODGE-01	Approved	Form submission persists record with ward metadata.
SCMS-F-006	File Evidence Upload	§ 4.2	FileUploadService	TC-LODGE-02	Approved	Restricts files to 5 MB; validates MIME types.
SCMS-F-007	Unique ID Generation	§ 4.2	ComplaintModule	TC-LODGE-03	Approved	Alphanumeric SCMS-YYYYMM-XXXXX format verified.
SCMS-F-008	Payload Field Validation	§ 4.2	ValidatorService	TC-LODGE-04	Approved	Blocks malformed and empty mandatory inputs.

Req ID	Requirement (Short)	Section Ref	Target Module	Test Case Ref	Status	Verification Notes
SCMS-F-009	Public Tracking Search	§ 4.3	TrackingModule	TC-TRACK-01	Approved	Public status retrieved without exposing PII.
SCMS-F-010	Citizen Dashboard	§ 4.3	UserPortal	TC-TRACK-02	Approved	Aggregates all user-specific active tickets.
SCMS-F-011	Audit Status History	§ 4.3	AuditLogger	TC-TRACK-03	Approved	Verified immutable status timeline entries.
SCMS-F-012	Citizen Append Notes	§ 4.3	ComplaintModule	TC-TRACK-04	Approved	Allows append before resolution; locked after.
SCMS-F-013	Admin Triage Filtering	§ 4.4	AdminDashboard	TC-ADMIN-01	Approved	Indexes yield fast response over 100k records.
SCMS-F-014	Department Assignment	§ 4.4	WorkflowModule	TC-ADMIN-02	Approved	Ticket routed and assignee notified.
SCMS-F-015	Resolution Verification	§ 4.4	WorkflowModule	TC-ADMIN-03	Approved	Rejects status transition if photo proof absent.
SCMS-F-016	Rejection Justification	§ 4.4	WorkflowModule	TC-ADMIN-04	Approved	Reason code stored and mailed to complainant.
SCMS-F-017	Automated Notifications	§ 4.5	NotificationQueue	TC-NOTIF-01	Approved	Async SMS and email delivery within 30s.
SCMS-F-018	Citizen Star Rating	§ 4.5	FeedbackModule	TC-NOTIF-02	Approved	1-5 rating recorded; token expires after 7 days.
SCMS-F-019	Analytics Data Export	§ 4.5	ReportingService	TC-REP-01	Approved	CSV/PDF reports stream cleanly.
SCMS-NF-001	Transaction Response ≤ 2s	§ 5.0	WebCore / Database	TC-PERF-01	Approved	Load testing verifies 90th-percentile response.
SCMS-NF-002	99.5% Service Uptime	§ 5.0	Infrastructure	TC-OPS-01	Approved	Health endpoints monitor continuous uptime.
SCMS-NF-003	500 Concurrent Capacity	§ 5.0	Infrastructure	TC-SCALE-01	Approved	Concurrency load validated without error spikes.
SCMS-NF-004	WCAG 2.1 AA Compliance	§ 5.0	Frontend UI	TC-UX-01	Approved	Lighthouse score verified ≥ 90.
SCMS-NF-005	Daily Backup Retention	§ 5.0	Storage Infra	TC-OPS-02	Approved	Verified backup snapshot script and restore.
SCMS-SR-001	TLS 1.3 Transport Security	§ 5.1.2	Web Server	TC-SEC-01	Approved	Port 80 to 443 redirect and TLS cipher suites.

Req ID	Requirement (Short)	Section Ref	Target Module	Test Case Ref	Status	Verification Notes
SCMS-SR-002	Argon2id/Bcrypt Hashing	§ 5.1.2	AuthModule	TC-SEC-02	Approved	Passwords stored with individual random salt.
SCMS-SR-003	Input Sanitization (SQLi/XSS)	§ 5.1.2	SecurityEngine	TC-SEC-03	Approved	Automated ZAP attack suite returns 0 vulnerabilities.
SCMS-SR-004	30-Minute Session Timeout	§ 5.1.2	SessionManager	TC-SEC-04	Approved	Token invalidation verified after 30m idle.
SCMS-SR-005	Magic-Byte File Whitelisting	§ 5.1.2	FileUploadService	TC-SEC-05	Approved	Rejects disguised executable payloads.